ASICONE (`asicone_202508`)

IHP SG13G2 OpenMPW September 2025 (T594): design, tapeout and silicon evaluation

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Abstract

ASICONE is a 1.4326 mm × 1.396 06 mm (2.0 mm^2) test chip fabricated in the IHP SG13G2 130 nm open-source PDK on the September 2025 OpenMPW shuttle (T594). It carries a generated 5-bit differential SAR ADC, two ring oscillators (13 and 101 stages) for reliability studies, and an SPI configuration block, all built with open-source tools and custom 18-track standard cells. This report collects the complete record of the chip: the blocks, the design flow, the tapeout sequence of August–September 2025 with its sign-off results and deviations, the adapter board and the bench, and the measurements compared with schematic and post-layout simulation.

Result in one paragraph. The chip is alive and every block responds. The ADC is monotonic with no missing codes at 1.43 MS/s, but reaches only 3.81 ENOB at low frequency (DNL −0.80 to 0.67, INL −0.87 to 0.99 LSB) because of binary-weight errors of −8 % to 7 % in its capacitive DAC; a post-layout simulation made after the fact reproduces the error pattern (INL correlation 0.98) and traces it to the wiring capacitance of the DAC plates, which could not be extracted at tapeout time. The SPI writes work, but the read-back (MISO) is dead because of a one-identifier typo in the RTL (`RDA[addr]` instead of `RDA[raddr]`), confirmed here by simulation. The 13-stage ring runs at 130.8 MHz, 31 % below the post-layout prediction; the 101-stage ring runs at 135.4 MHz, almost five times *faster* than predicted, which no parasitic can explain and which remains an open functional anomaly.

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1 Summary of results

Block	Status	Key numbers (bench)
SAR ADC, 5 bit	Works, poor linearity	1.4286 MS/s; no missing codes; ENOB 3.81 at 1 kHz, 3.08 at 650 kHz DNL −0.80 to 0.67 LSB, INL −0.87 to 0.99 LSB (0.51 LSB rms)
RO13	Oscillates, slow	130.8 MHz (post-layout TT 188.8 MHz, SS 169.3 MHz)
RO101	Oscillates, anomalous	135.4 MHz (post-layout TT 27.7 MHz): 4.9× too fast
SPI, write	Works	RO configuration written and obeyed (16-value scan)
SPI, read (MISO)	Broken	RTL typo `RDA[addr]` → `RDA[raddr]`; reads return the undriven register 0
I/O ring, ESD, supplies	Works	38 bonded pads; core 1.2 V, I/O 3.3 V

Table 1: Status of every block of ASICONE.

Item	Value
Process	IHP SG13G2 (130 nm SiGe BiCMOS, used as SG13CMOS), open-source PDK
Shuttle	MPW 0.13 µm September 2025 (T594), 10 samples, final wafer thickness 200 µm
Top cell / GDS	`asicone_202508` / `asicone_202508.gds` (with fill)
Die (incl. seal ring)	1432.60 µm × 1396.06 µm = 2.000 mm^2 (the full allotment)
Pad-ring area	1160 µm × 1122 µm (inside seal ring, guard and bond pads)
Pads	38 bonded (`sg13g2` I/O library, 70 µm bond pads), QFN48 package
Supplies	VDD 1.2 V (core), AVDD 1.2 V (ADC), ROVDD/RO2VDD 1.2 V (rings), VDDIO 3.3 V
Standard cells	Custom 18-track `sg13g2` (6.12 µm rows, 86 cells) and floating-body `sg13g2f` (77 cells, ADC)
Tools	Yosys, OpenROAD (custom scripts, no ORFS), KLayout DRC/LVS, ngspice, Xyce, iverilog

Table 2: Chip data.

2 The chip

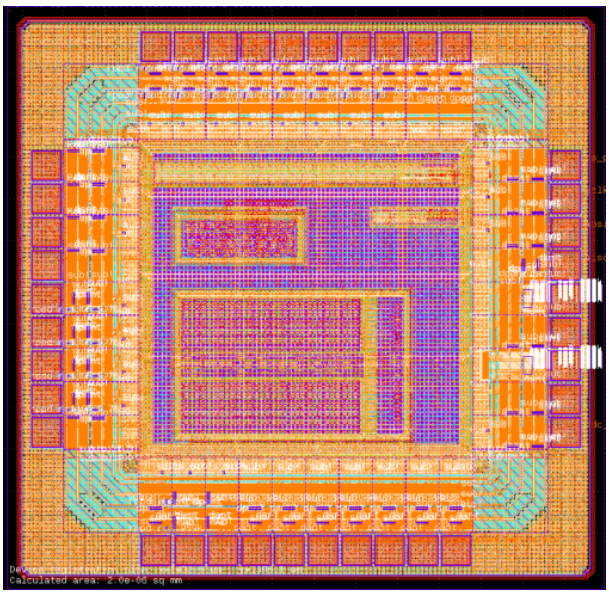
2.1 Contents and floorplan

ASICONE was conceived as a vehicle to evaluate the open-source flow and the IHP SG13G2 PDK on real silicon, not as a product. It contains:

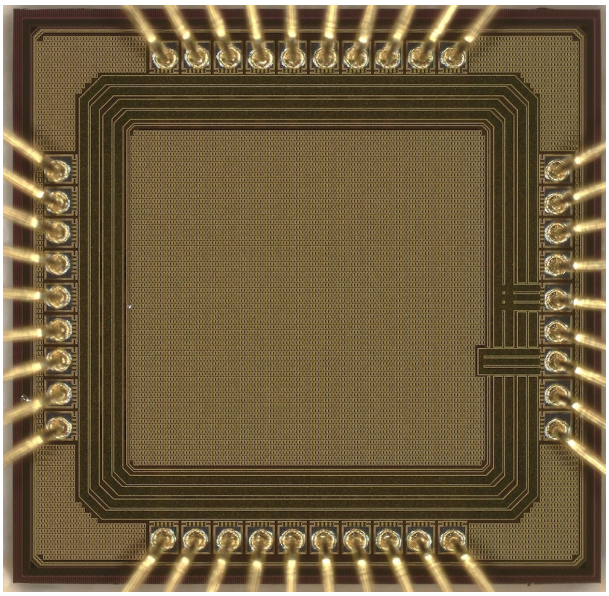
- **SARADC**: a 5-bit fully differential charge-redistribution SAR ADC produced by the `saradc_auto` generator [1]: MOS-gate unit capacitors, a custom switch cell, a dynamic comparator and synthesized SAR logic, all placed and routed by OpenROAD.
- **RO13 and RO101**: two ring oscillators (`ro_reliability`, cells `top_13` and `top_101`) with a device under test (DUT) stage that can be isolated for stress and I–V measurement, each followed by a ÷32 divider and an output pad.
- **SPI**: a configuration slave (8 write registers, 6 read registers of 8 bits) whose registers drive the RO control bits.

Instance	Cell	Origin (µm)	Size (µm)
`adc`	`SARADC`	(265.0, 220.0)	566.44 × 361.08
`spi`	`SPI`	(265.0, 643.0)	311.62 × 128.35
`ro_101`	`top_101`	(216.0, 838.0)	727.77 × 48.76
`ro_13`	`top_13`	(738.6, 739.54)	205.17 × 31.81

Table 3: Macros of the top level (coordinates of the 1160 µm × 1122 µm pad-ring area, from `chip/pnr/outputs/asicone_202508.def`). Seven `BUFFD1` and tie cells at the top level build the read-back word of the SPI.



(a) Layout (KLayout capture of the submitted GDS).



(b) Die photograph, bonded in the QFN48 package.

Figure 1: ASICONE. In the photograph the metal fill hides the circuits; the 38 bond wires (10 on the top and bottom edges, 9 on the sides) are visible, and so is the seal ring.

2.2 Pinout

The die is bonded in a QFN48 (0.5 mm pitch); 10 package pins are left open. Table 4 is the pinout as connected on the adapter board (Section 5.2).

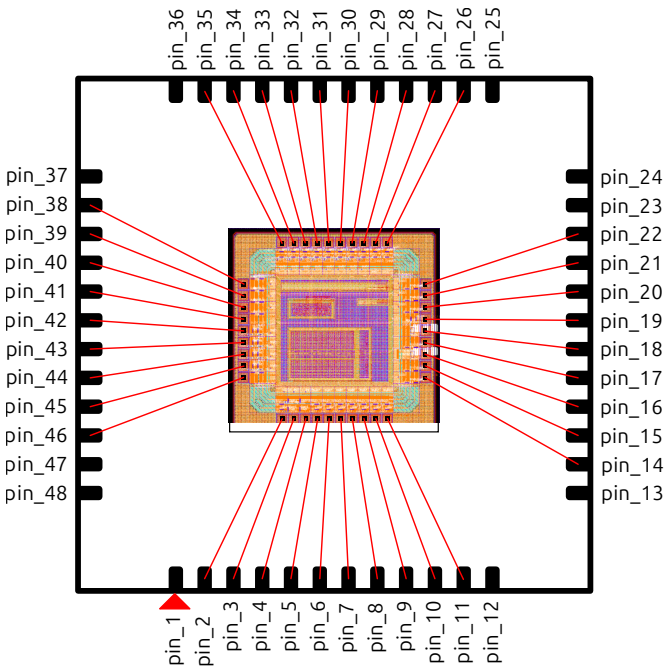


Figure 2: Bonding diagram (`QFN48.svg`).

Pin	Signal	Pad cell	Board net	Pin	Signal	Pad cell	Board net
2	AVDD	IOPadAnalog	`adc_vdd` (JP5)	26	RO_13_Vout	IOPadOut16mA	`ro_13_vout`
3	adc_rst	IOPadIn	`adc_rst`	27	RO_101_Vout	IOPadOut16mA	`ro_101_vout`
4	adc_clk	IOPadIn	`adc_clk`	28	RO2VDD	IOPadAnalog	`ro2_vdd` (JP4)
5–9	adc_result[0..4]	IOPadOut16mA	`adc_result_0..4`	29	ROVDD	IOPadAnalog	`ro_vdd` (JP3)
10	adc_valid	IOPadOut16mA	`adc_valid`	30	RO_13_extra_load	IOPadAnalog	`ro_13_ex`
11	adc_sample	IOPadOut16mA	`adc_sample`	31	RO_13_Drain_Sense	IOPadAnalog	`ro_13_ds`
14	adc_go	IOPadIn	`adc_go`	32	RO_13_Drain_Force	IOPadAnalog	`ro_13_df`
15	VSS	IOPadVssExt	GND	33	RO_101_extra_load	IOPadAnalog	`ro_101_ex`
16	VDD	IOPadVddExt	`vdd1p2` (JP2)	34	RO_101_Drain_Sense	IOPadAnalog	`ro_101_ds`
17	VSSIO	IOPadIOVss	GND	35	RO_101_Drain_Force	IOPadAnalog	`ro_101_df`
18	VDDIO	IOPadIOVdd	`vdd3p3` (JP1)	38	RO_13_DUT_gate	IOPadIn	`ro_13_gate`
19	miso	IOPadOut16mA	`miso`	39	RO_101_DUT_gate	IOPadIn	`ro_101_gate`
20	mosi	IOPadIn	`mosi`	40	RO_RST_B	IOPadIn	`ro_rstb`
21	sclk	IOPadIn	`sclk`	41	VSS (RO)	IOPadVssExt	GND
22	cs	IOPadIn	`cs`	42	adc_vip	IOPadAnalog	`adc_vip` (JP9)
				43	adc_vin	IOPadAnalog	`adc_vin` (JP10)
				44	adc_vrefn	IOPadAnalog	`adc_vrefn` (JP7)
				45	adc_vrefp	IOPadAnalog	`adc_vrefp` (JP6)
				46	VSS (ADC)	IOPadVssExt	GND

Table 4: Pinout. Open package pins: 1, 12, 13, 23, 24, 25, 36, 37, 47, 48. Analog signals connect to the core through the `padres` terminal of `sg13g2_IOPadAnalog` (secondary protection resistor of about 587 Ω and ESD diodes); the ring supplies use the direct `pad` terminal.

3 Block design

3.1 Custom standard cells

Every digital cell on the chip is custom. Two libraries were drawn for this tapeout:

- `sg13g2` (`lib/`): 86 cells, 18 tracks (6.12 µm rows on a 0.34 µm site), used by the SPI and the top level;
- `sg13g2f` (`saradc_auto/cells/`): 77 floating-body variants used inside the ADC, plus the analog cells `SARADC_CELL_INVX16_ASCAP` (unit capacitor) and `SARADC_CELL_INVX0_ASSW` (switch).

The 18-track height is generous and the cells are not optimized: the standard-cell generator was still in development. The Liberty timing was *not* characterized in SG13G2; it was adapted from `sky130`, because the PDK had no parasitic extraction to characterize the layouts with. Timing of the digital blocks is therefore nominal only; at the clock rates used (10 MHz for the ADC, a few MHz for the SPI) this was never a limitation.

3.2 SAR ADC

Architecture. The ADC is a fully differential, top-plate-sampling charge-redistribution SAR ADC. Each side has a binary capacitive DAC of 2^5 units (bits of 1, 2, 4, 8, 16 units plus a dummy) between a switched plate (VSH: input, VREFH or VREFL) and a floating top plate (FL) that feeds the comparator inputs VOUTH/VOUTL. The unit capacitor is the gate of a group of MOS transistors (67.8 fF per unit with FL floating); there are 720 unit-capacitor cells and 1452 switch cells. The comparator is a dynamic latch built from `sg13g2f` cells; the SAR logic (`sar_logic_wreset.v`) is synthesized. A conversion takes $N + 2 = 7$ clocks: sampling, five bit decisions and the result. Interface: `CLK`, `RST` (active high), `GO` (start conversions), `SAMPLE`, `VALID` and `RESULT[4:0]`.

Generator and flow. `saradc_auto` produces the ADC from RTL: Yosys synthesizes the analog netlist (capacitors and switches as cells) and the logic; OpenROAD places the DAC array with a scripted pattern (a "circle" pattern with Metal5/TopMetal1 bridges for bits 1 and 3), routes the special nets (plates, references, top plates) with custom Tcl, and routes the rest with the detailed router. The taped-out version used the flat flow now kept as `pnr_legacy/`. Macro size 566.44 µm × 361.08 µm, 24 198 transistors. The taped-out netlists, DEF, GDS and reports are archived in the release `tapeout-T594` of `saradc_auto` (`https://github.com/ckdur/saradc_auto/releases/tag/tapeout-T594`); its `SARADC.gds` is identical, layer by layer, to the `SARADC` cell of the submitted chip GDS.

Design-time simulation. Before tapeout the ADC was verified at schematic level with ngspice and Xyce at 1.8 V: transient conversions (only a few samples were possible at the time with the analog netlist), multi-frequency tests, a capacitor-linearity test and a switch-charge test (commits of 30 August to 5 September 2025). No layout parasitics were available (Section 4.4).

3.3 Ring oscillators

Each ring (`top_13`, `top_101`) has 13 or 101 inverting stages of `lv` MOS ($L = 0.45$ µm, $W_p/W_n = 4\times3.4$ µm$/4\times1.3$ µm), a 5-flop ripple divider (÷32, reset by `RO_RST_B`) and a 16 mA output pad. Three stages are special: `rovcel3` closes or opens the ring (header/footer on `n_RO_control`, `RO_control`) and, with the ring open, connects `DUT_gate` to the DUT input; `rovcel1` is the DUT, with a PMOS header (`DUT_Header`) and NMOS footer (`DUT_Footer`), whose output (`extra_load`) goes straight to an analog pad and, with the ring open, to `Drain_Force`/`Drain_Sense` through transmission gates; `rovcel4` has a PMOS header on `n_RO_control`. RO101 is supplied from ROVDD, RO13 from RO2VDD; both share the core VSS. The control bits come from SPI register 0 (RO101, `R[3:0]`) and register 1 (RO13, `R[11:8]`): bit 0 `n_RO_control`, 1 `RO_control`, 2 `DUT_Footer`, 3 `DUT_Header`; oscillation is `0x06`. The ROs were integrated last, on 12 September 2025, the day of the final submission.

3.4 SPI

`digital/rtl/spix/spi.v`, parameters `registers=8`, `inputs=6`, `sword=8`, `syncing=0` (one-hot shift counter), `addressing=1` (multiplexer read). Frame, sampled on the rising SCLK with CS low: write bit, read bit, `addr[2:0]`, `data[7:0]`, then, for a read, 8 bits shifted out on MISO. The register is written on the 14th edge; the bit counter only restarts on an SCLK edge with CS high. Synthesized to 108 `DFCNQD1` flops, 91 `MUX2D1` and about 140 other gates; macro 311.6 µm × 128.4 µm. Read map: register 0 = ADC result, 1 = {sample, valid}, 2/3 = constant `0xAA`/`0x55`, 4/5 = write registers 0/1 (echo). The SPI reset is the `adc_rst` pad (Section 8).

4 Design flow and tapeout sequence

4.1 Flow

The flow is custom (no OpenROAD-flow-scripts), organized per block as `syn` (Yosys) → `pnr` (OpenROAD) → `signoff` (KLayout GDS merge, fill, DRC, LVS), and at the top as `padring` (the `padring` tool, `.cfg` with corners, pads, bond pads and fillers) → `pnr` (macro placement, power grid, routing) → `signoff`. Versions used for the final sign-off:

IHP-Open-PDK	`6bda9f1cd9ae7108571041d9bd0f03a5a94f731f`
OpenROAD	`59ef9a94573c0a6799d32e2dfc9fc7a93feeb30f`
Yosys	`4581f75b03a25b615a9715516854c424b580f7ab` (0.49 for the SPI netlist)
KLayout	0.30.2 (`d1dc885235bf571eac7270d6dccca820a019b4b1`)
ngspice	44.2
OpenVAF-Reloaded	`802bce21ac44833b52c010cb90d37e0d6fddd4ac`

4.2 Timeline

Reconstructed from the commit history of `asicone_sg13g2`, of the submission fork `TO_Sep2025` and of the sign-off logs (all times JST).

Date	Step
Preparation (May–July 2025)	
2025-05-12	First commit: digital flow for the custom cells.
05-16 – 05-19	First OpenROAD PnR of a digital block, up to CTS.
06-16 – 06-30	First full PnR and sign-off of the digital block; DRC working, LVS not yet.
07-21	Synthesis fixed, first LVS-clean digital block; fillers added and first pad ring generated; first top-level PnR draft.
07-23 – 07-29	Pad-ring configuration, first OpenROAD script for the chip; the ADC brought into the repository.
ADC physical design (August 2025)	
08-05 – 08-08	First synthesis of the ADC, placement, blockages; first ADC with all routes but no connections.
08-12	IHP opens the `TO_Sep2025` repository (rules: 2 mm^2 including seal ring, registration 1 September, DRC-clean GDS by 14 September, no changes afterwards).
08-14	Via generation for the custom routes.
08-17 – 08-19	Joining of the ADC routes, power stripes, first placement and CTS of the chip.
08-20	First complete PnR; final pad-ring and seal-ring size; bond pads. Placeholder submitted to `TO_Sep2025`.
08-21 – 08-22	Assets reorganized, LEF fixes, comparator routes; full chip flow complete (unverified), better detailed routing.
DRC and LVS closure (23–28 August)	
08-23	DRC fixes in the metal stack and in the standard cells.
08-24	ADC DRC reduced from about 5000 to 20 violations, then 38 left at the top.
08-25	ADC DRC clean; LVS debugging, cells rejected by LVS removed.
08-26	LVS clean on the SARADC; libraries renamed and separated (`sg13g2`/`sg13g2f`); full chip assembled.
08-27	Manufacturing-grid fixes, DRC fixes, custom LVS deck support; die size corrected for the seal-ring-to-passivation rule (`TO_Sep2025`).
08-28	I/O cells customized for LVS (missing `PolyRes` recognition layer added); `ptap1` ignored in a custom deck; **full-chip LVS pass**.
Verification and submission (29 August – 14 September)	
08-29 – 08-31	ADC testbenches; submission metadata; simulation folder of the submission.
09-01	Registration deadline. ADC simulations ported to Xyce; SAR logic changed to flops with reset for the result.
09-02 – 09-05	Multi-frequency, capacitor-linearity and switch-charge simulations of the ADC.
09-09	Correction of the `sg13g2` library.
09-12 01:13	First integration of the ring oscillators into the chip.
09-12 10:18	First sign-off of the chip with the ROs.
09-12 21:18	Final sign-off runs on `asicone_202508.fill.gds`: LVS (KLayout 0.30.2, flat, custom deck) "Netlists match" at 22:53 (95 min); full DRC with antenna and density at 23:15 (117 min).
09-12 23:40	Final upload to `TO_Sep2025` (`asicone_202509`): GDS, DRC, LVS, simulation and digital data, metadata.
09-12 23:49	"Everything ready" in `asicone_sg13g2`.
09-14	Shuttle deadline.
After the tapeout	

Date	Step
09-15 – 10-03	Submodule fixes, pad-ring tool included, license and README; the ADC moved to its own repository (`saradc_auto`).
2025-12 – 2026-01	Repositories moved to GitLab, `run.sh` demo, fixes for newer OpenROAD.
2026-05-08	QFN48 bonding diagram.
2026-08-17	Die photograph of the packaged chip.
2026-08-28 – 09-24	Adapter board designed (first commit; fabrication files on 09-24) (`IHP130_ASICONE_ADP_QFN48_0p5Q_Adapter`).
2026-09-12 – 09-25	RISCVConsole SoC ported to the ULX3S (SD boot, extra SPIs, external 10 MHz clock).
2026-10-01 – 10-19	First ADC conversions on the bench, fast sampling firmware.
2026-10-22	ADC sweep with the MSO20 (data set `sweep_20261022`).
2026-10-24 – 10-25	ADC compared with schematic and post-layout simulation.
2026-10-29 – 10-30	Ring-oscillator measurement (`ro_20261029`) and post-layout comparison.

The critical path was clear: the ADC went from its first routed layout (8 August) to DRC clean (25 August) and LVS clean (26 August) in under three weeks, leaving two weeks for the full-chip verification, the simulations and the RO integration, and no time for any post-layout iteration.

4.3 Sign-off results and deviations

- **DRC**: the full KLayout deck (flat, antenna, density) reports 52 violations, all in the IHP I/O cells: 26 `Sdiod.d` and 26 `Sdiod.e`. They were accepted; the I/O works on silicon. All the designed blocks are DRC clean.
- **LVS**: "Netlists match" with a custom deck: `ptap1` is ignored entirely (the extractor turned substrate taps into diodes, which only match if each tap is a separate diode, and merged taps in the layout create a single diode), and a `PolyRes` layer missing in some I/O sub-cells was added so that the `rppd` resistors are recognized.
- Capacitors were attached to some transistors of the netlist for simulator convergence.
- No parasitic extraction (Section 4.4) and no timing characterization in SG13G2.
- No gate-level or RTL regression of the SPI read path: its testbench (`spi_tb.v`) still used an old port name (`DOUT`) and no longer elaborated against the taped-out `spi.v`. DRC and LVS only check that the layout implements the netlist; they cannot catch a logic error in the RTL (Section 8).

4.4 Parasitic extraction: not available at tapeout

At tapeout the SG13G2 PDK had no parasitic-extraction rules, so neither the ADC nor the ROs could be simulated with their layout. After the measurements several routes were tried: Magic 8.3.623 (capacitances about 100× too large, and Netgen cannot compare the netlist), and KLayout-PEX [6] on the whole flat layout (single-threaded 2.5D engine; runs of 4, 12, 10 and 16 h never finished). What finally worked is a *split* extraction: the layout is cut into small windows along fillers and taps (81 for the ADC), each window is extracted with KLayout-PEX 0.3.12 (2.5D, coupling capacitances, *no resistance*) after the PDK LVS deck with its unconditional `purge` disabled, and the windows are stitched by net name. The stitched ADC netlist has the 24 198 transistors and 68 519 capacitors and matches the schematic on all 572 named nets. The RO cells were extracted the same way from the fabricated GDS. These are the post-layout results quoted in this report; they were not available when the chip was designed.

5 Measurement environment

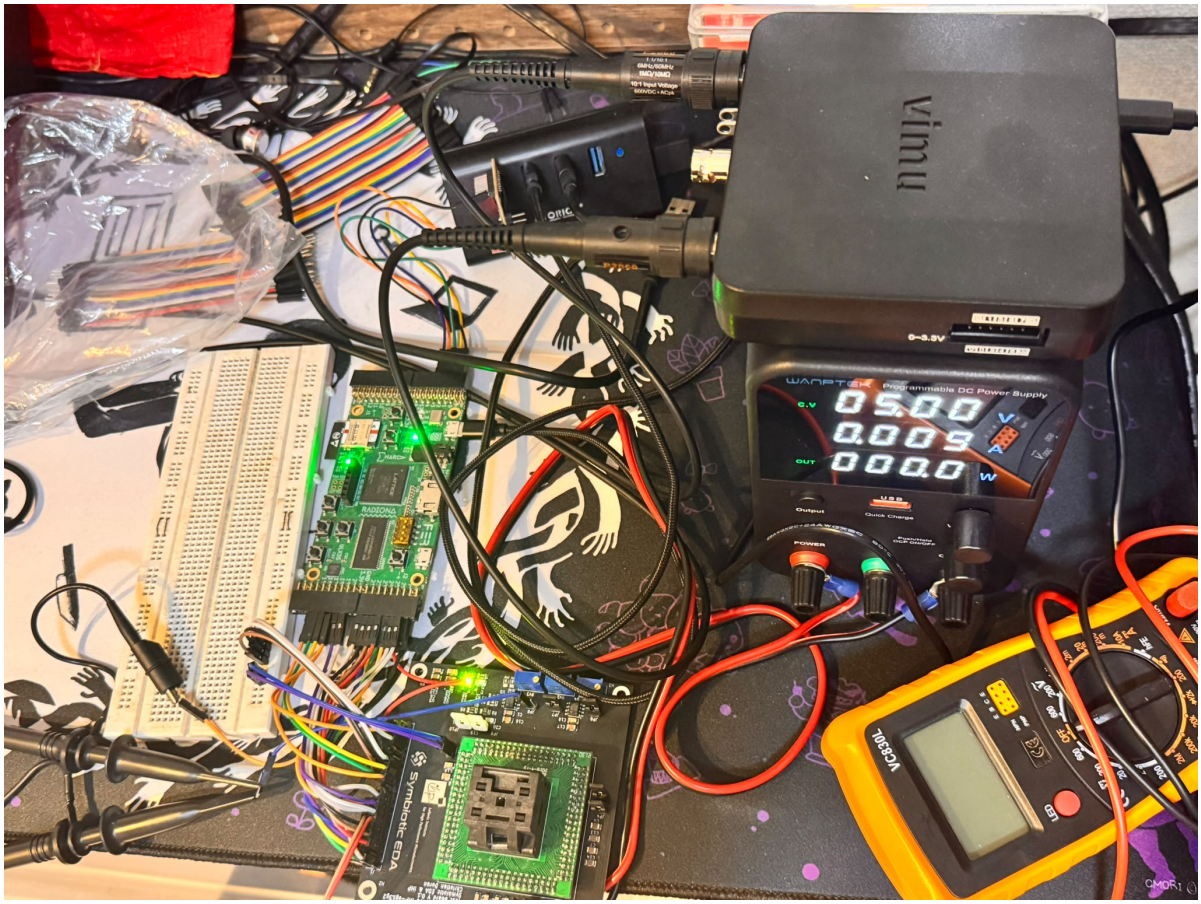


Figure 3: The bench. Bottom centre: the adapter board with the chip in a QFN48 clamshell socket; left of it the ULX3S FPGA board (Lattice ECP5) wired to the adapter header J1 with jumper wires and a breadboard; top right: the Vimu MSO20 oscilloscope/DDS with its probes; right: a Wanptek programmable DC supply and a VC830L multimeter (used for the reference and supply voltages); a powered USB hub connects the instruments, the JTAG adapter and the FPGA to the host PC.

5.1 Equipment and signal chain

Device under test	ASICONE in a QFN48 clamshell socket on the adapter board
Adapter board	`IHP130_ASICONE_ADP_QFN48_0p5Q_Adapter` (Section 5.2)
Digital host	ULX3S, Lattice ECP5-85F, RISCVConsole SoC: RV64IMAC Rocket core, buses at 50 MHz; ADCFIFO (`0x64003000`), SiFive SPI (`0x64002000`), `rocounter` (`0x64005000`)
Firmware	RISCVConsoleCode, branch `ihp` (`src/adc.c`, `src/ro.c`)
Signal source	Vimu MSO20BL, DDS channel 1, driven by VmMsoOpenSDK (`software/mso/mso_dds`)
Voltage reference	MSO20 channel 1 on the DDS output, DC, ±1 V range, 8192 samples per capture
ADC clock	10 MHz from the FPGA PLL; 7 clocks per conversion, $f_s = 1.4286$ MS/s
Frequency counting	`rocounter`: 3-flop synchronizer and edge counter, gate of 50 000 000 bus cycles (1 s, 1 Hz resolution at the pad, 32 Hz at the ring) or 100 ms
Debug and data	FT232H JTAG (OpenOCD) for boot and buffer dump; ULX3S USB UART at 115 200 baud
DC	Multimeter VC830L; bench supply (Wanptek) available for supply sweeps through JP3/JP4

Table 6: Equipment.

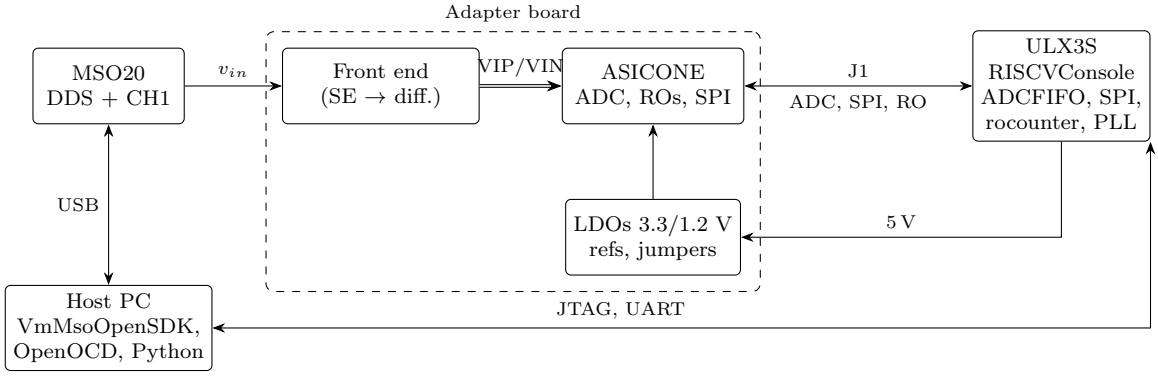


Figure 4: Signal chain of the bench.

5.2 Adapter board

A two-layer 85.5 mm × 85.5 mm board (KiCad, `pcbs/IHP130_ASICONE_ADP_QFN48_0p5Q_Adapter`, assembled from the JLCPCB BOM/CPL files) that plugs into the ULX3S 2×20 header and carries everything the chip needs. All passives are 1206.

Ref.	Part	Function
U1	QFN48 socket	ASICONE
U2	TLV75533PDBV	3.3 V LDO from the 5 V of J1: op-amps and chip I/O (VDDIO via JP1)
U3	TLV75512PDBV	1.2 V LDO: core (JP2), ADC (JP5), ROVDD (JP3), RO2VDD (JP4), references, front-end divider
U4	LMV358	Buffers of VREFP (U4A, JP6) and VREFN (U4B, JP7), set by RV1/RV2
U5	LMV358	Single-ended to differential front end (SBOA216 [5])
RV1–RV3	10 kΩ Bourns 3296W	VREFP, VREFN, and a DC input (RV3, through JP8)
R1, R2	10 kΩ	0.6 V common-mode divider
R3, R4	1 kΩ	Gain resistors of the inverting leg (U5B)
R5, R6	47.5 Ω	Isolation resistors to the ADC inputs
C19 / C20, C21	1.2 nF / 120 pF	Differential / common-mode charge reservoir at VIP/VIN
C1–C18	10 µF + 100 nF	Decoupling of every supply pin
Q1, Q2, D1, D2, R7, R8	2N7002, LED, 220 Ω	"1.2 V" and "3.3 V" power indicators (MOSFET gate on the LDO output)
JP1–JP10	2-pin jumpers	Supply, reference and input selection; current measurement points
J1	2×20 header	ULX3S connection

Table 7: Adapter board components.

Front end. U5A buffers the input ($v_p = v_{in}$); U5B is a difference amplifier with R3/R4 referenced to the 0.6 V of R1/R2 ($v_n = 1.2\,\mathrm{V} - v_{in}$), so the ADC sees $v_d = 2(v_{in} - 0.6\,\mathrm{V})$ at a 0.6 V common mode. R5/R6 with C19–C21 form a differential pole at 1.33 MHz and a common-mode pole at 28 MHz. With VREFP/VREFN = 0.8 V/0.4 V the ideal LSB is 25 mV differential, 12.5 mV at the board input.

Header J1. Pins 1–17: `adc_go` (1), `miso` (2), `adc_sample` (3), `mosi` (4), `adc_valid` (5), `sclk` (6), `adc_result_4..0` (7, 9, 13, 14, 15), `cs` (8), 5 V (11), GND (12, 30), `adc_clk` (16), `adc_rst` (17). RO signals: `ro_rstb` (23), `ro_101_gate` (24), `ro_13_gate` (25), `ro_101_df/ds/ex` (26/27/28), `ro_13_df/ds/ex` (31/32/33), `ro_101_vout` (35), `ro_13_vout` (36).

Known limitations of the board. (1) The LMV358 (GBW about 1.4 MHz fitted, slew rate about 1 V/µs) limits the front end above 100 kHz, and its two legs have different bandwidths (unity gain vs. noise gain 2), which turns signal into common mode. (2) The references have no decoupling capacitor at the chip pins. (3) The analog RO pins (`df`, `ds`, `ex`) go to J1 directly and must never see the 3.3 V of the FPGA. A revised front end (fully differential amplifier, decoupled references) is proposed in `RISCVConsole/software/mso/pcbfix`.

6 SAR ADC measurement

Conditions: VDD = AVDD = 1.2 V, VREFP/VREFN = 0.8 V/0.4 V (multimeter), input common mode 0.6 V, 10 MHz clock, $f_s = 1.4286$ MS/s, 65 536 samples per record, room temperature. Static linearity by the code-density test of IEEE 1241 [3] with a 218 Hz triangle overdriving both ends; dynamic performance with coherent sines (odd prime number of periods) and the four-parameter fit of IEEE 1057 [4]. All voltages are single-ended at the board input. The full analysis is in `RISCVConsole/software/mso/results/sweep_20261022/report/report.pdf`.

6.1 Static linearity

Input range (first to last transition)	0.3719 V to 0.7696 V (ideal 0.4125 V to 0.7875 V)
LSB	13.26 mV (ideal 12.50 mV; gain error 6.1 %)
Offset	−29 mV (−2.2 LSB), board and comparator
DNL	−0.80 to 0.67 LSB
INL (end point)	−0.87 to 0.99 LSB, 0.51 LSB rms
Missing codes	none; monotonic

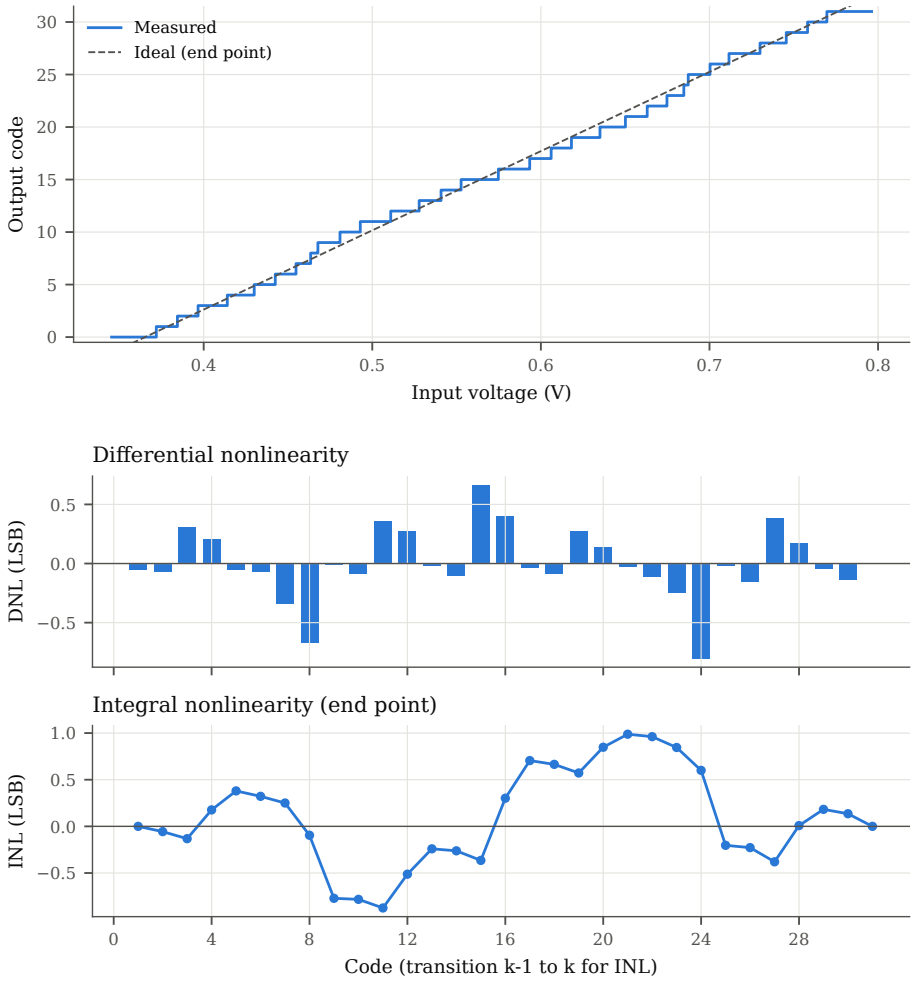


Figure 5: Transfer characteristic and DNL/INL from the histogram test.

The DNL repeats every 8 codes and the INL jumps at codes 8, 16 and 24: the signature of wrong binary weights in the DAC. A least-squares fit of the transition levels to the bits gives:

Bit	0	1	2	3	4
Ideal weight (LSB)	1	2	4	8	16
Measured weight	0.92	1.89	4.27	7.33	16.55
Measured error (%)	−7.8	−5.7	+6.7	−8.4	+3.5
Schematic simulation (%)	+0.0	−0.1	+0.0	+0.0	+0.0
Post-layout, taped-out GDS (%)	−12.5	−6.8	+3.3	−5.2	+2.3

Table 8: Binary weights of the DAC. After removing them, 0.19 LSB rms of INL remain.

6.2 Dynamic performance

f_{in} (kHz)	A (LSB)	SINAD (dB)	ENOB	THD (dB)	SFDR (dB)	H2 (dBc)	H3 (dBc)
1.0	13.29	24.69	3.81	−27.5	32.9	−45.4	−31.5
10.0	13.29	24.85	3.84	−27.4	33.0	−45.1	−31.4
50.0	13.27	24.73	3.82	−27.4	32.4	−46.2	−32.5
100.1	13.09	23.93	3.68	−26.0	29.9	−43.9	−34.6
200.0	12.80	23.48	3.61	−25.2	29.8	−43.9	−31.1
300.0	12.16	23.37	3.59	−25.3	29.9	−39.0	−33.2
500.0	9.89	22.29	3.41	−24.3	30.7	−27.5	−32.2
650.0	6.66	20.30	3.08	−22.9	27.6	−24.3	−29.9

Table 9: Sine sweep (input 360 mV peak-to-peak DDS setting, 565 mV bias). An ideal 5-bit quantizer at the low-frequency amplitude gives 30.2 dB (4.72 bits).

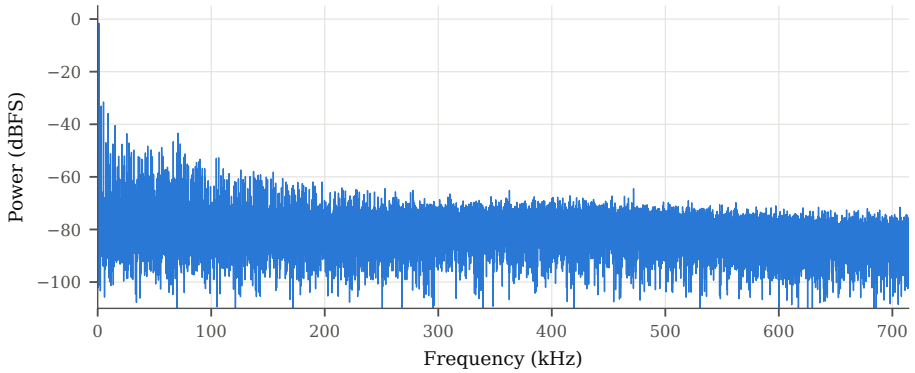


Figure 6: Spectrum at 1 kHz.

Below 50 kHz the measured SINAD equals, within 0.3 dB, the SINAD predicted from the ramp transfer alone: the ENOB of 3.81 bits is entirely the static nonlinearity, i.e. the DAC weights. Above 100 kHz the board front end takes over: the amplitude falls to half at 650 kHz, the common-mode ripple reaches 47 % of the signal and the op-amps approach their slew limit, which produces the rising H2.

6.3 Comparison with simulation

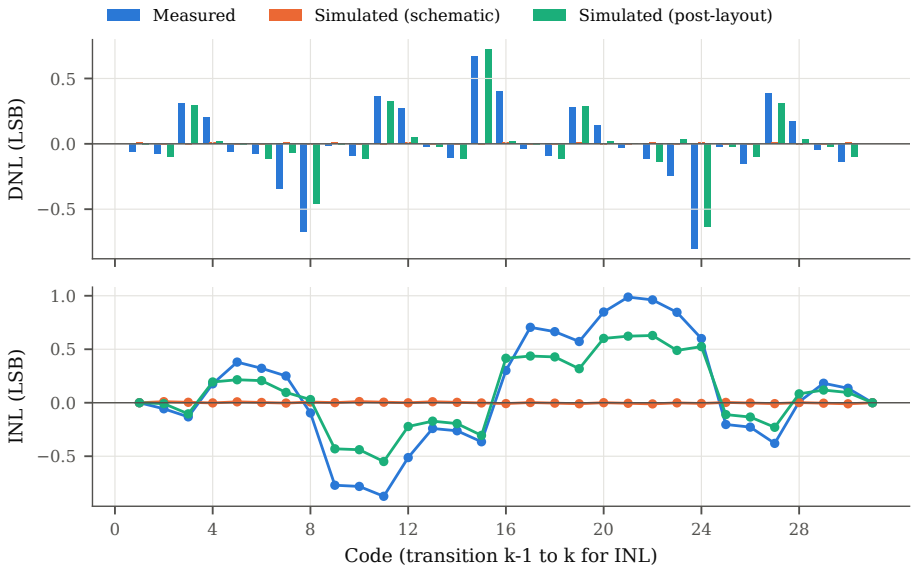


Figure 7: DNL/INL: chip, schematic and post-layout simulation of the taped-out layout.

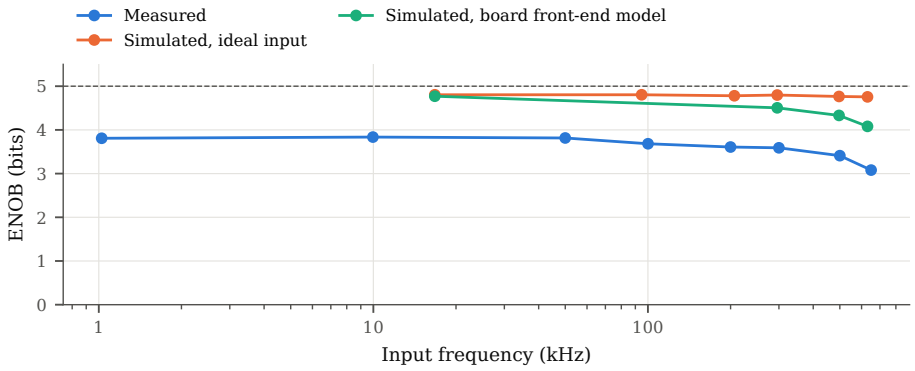


Figure 8: ENOB: chip, transient simulation with ideal input, and with the front-end model of the board.

- **Schematic** (ngspice-47, PSP103, TT, 25 °C, board conditions): ideal — weights within 0.1 %, INL 0.006 LSB rms, ENOB 4.80 at low frequency and 4.76 at 631 kHz. The converter core has no dynamic loss up to Nyquist.
- **Post-layout** (split extraction of the taped-out GDS): weight errors with the sign of the measurement on all five bits, INL 0.33 LSB rms, INL correlation 0.98 and DNL correlation 0.90 with the chip, gain error 7.1 % (measured 6.1 %). Quantizing the measured input with the post-layout transitions gives 4.19 bits: the layout accounts for about half of the lost bit, and for the whole error *pattern*.
- **With the board model**, the simulated ENOB falls by about 0.7 bits up to 631 kHz, the same loss as measured (3.81 → 3.08); the measured curve is about one bit below the simulated one at every frequency — the static weights again.

6.4 Why the ADC performance is poor

The "efficiency" of the ADC — how much of its 5 bits it delivers — is low: 3.8 of the 4.7 bits an ideal quantizer would give at the same amplitude, and less at high frequency. The causes, in order of weight:

1. **DAC plate wiring capacitance (layout).** The plates of each bit have wiring capacitance of the order of the unit capacitor itself (about 100 fF per unit on VSH, 65–75 fF on FL), and part of it couples

directly to the comparator inputs, differently for every bit: the bit 1/3 bridges add about 10 % of wiring capacitance; the MSB switched plate couples 11.3 fF per unit to its own comparator input, the others about 5 fF; the floating plates of bits 0–3 couple 7–10 fF per unit to the opposite input; and the two halves of the flat layout are not mirror images. This is fully determined by the generator's routing, not by the process.

2. **No extraction at tapeout and no time to iterate.** Without extraction rules for SG13G2 the layout could only be verified for DRC and LVS. The ADC reached LVS clean 17 days before the deadline, and the rest of that time went into full-chip closure. The effect was invisible until the split extraction of October 2026.

3. **Remaining, unexplained part** (0.20 LSB rms of INL; bit 0 less reduced, bit 2 larger, bit 3 smaller than post-layout): of the size that random unit-capacitor mismatch, the wiring resistance (not extracted) or the undecoupled board references can produce. A ramp with 100 nF + 1 µF at pins 44/45 would separate board from chip.

4. **The board** above 100 kHz (LMV358 front end), which is not the ADC.

The generator has since been corrected (branch `dac-routing-fix`, `docs/dac_wiring.md`): a bridge-less nested capacitor pattern and re-assigned switch-row rails bring the per-unit plate capacitance of all bits within a few percent and the plate-to-comparator coupling below 0.5 fF; a hierarchical layout with one half mirrored halves the post-layout INL (0.17 LSB rms, 4.52 bits). The power of the ADC was not measured, so no figure of merit is given; the current through JP5 is the next measurement.

7 Ring oscillator measurement

Conditions: ROVDD and RO2VDD from the on-board 1.2 V LDO through JP3/JP4, room temperature, `df`/`ds` tied to 1.2 V, `ex` open, `adc_rst` held high (it is the SPI reset). Firmware `ro_test`, `ro_diag`, `ro_scan`; five runs, 132 restarts per ring in six different ways. Full report: `RISCVConsole/software/ro/results/ro_20261029/report/report.pdf`.

7.1 Results

	RO13	RO101
Pad, ring alone (1 s gate)	4.088 MHz	4.231 MHz
Ring frequency (pad × 32), alone	130.8 MHz	135.4 MHz
Change when the other ring runs	−4.0 %	0.2 %
Repeatability (std. dev.)	0.10 %	0.11 %
Spread over 132 restarts	1.24 %	0.24 %
Duty cycle at the pad	49.6 %	49.7 %
Schematic + pad, TT	223.9 MHz	32.3 MHz
Post-layout (wiring C), TT / SS / FF	188.8 / 169.3 / 209.8 MHz	27.7 / 25.0 / 30.6 MHz
Post-layout, TT, 1.0 V	139.5 MHz	20.7 MHz
Measured / post-layout TT	0.69	4.88
f_{13}/f_{101}	measured 0.97, predicted 6.93 (6.86–6.99 over corners)	

Table 10: Ring oscillators: bench against simulation.

Function. The SPI writes arrive: in the 16-value scan the rings oscillate only for `0x06`/`0x0e` (`RO_control`=1, `n_RO_control`=0, `DUT_Footer`=1), `RO_RST_B` low stops both outputs and the 50 % duty cycle confirms the dividers. Two observations do not match the design: `DUT_Header` (bit 3) has no visible effect — with the header off the rings keep running (RO13 4.5 %, RO101 −0.2 %) where the DUT can only pull down and the ring should stop — and with the ring open and `DUT_gate`=1 (`0x04`, `0x0c`) RO13 shows a slow 12 kHz oscillation at the pad.

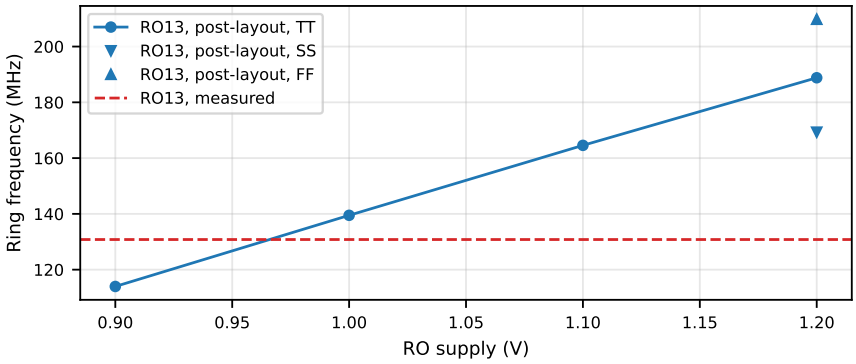


Figure 9: Post-layout RO13 against its supply, and the measured frequency.

7.2 Is the mismatch justified?

The ADC story — missing extraction and no time to iterate — explains a *slower* circuit, because wiring capacitance and resistance can only lower the frequency. It explains part of RO13 and nothing of RO101.

RO13: partly justified, not fully. The extraction lowers RO13 by 16 %, but the chip is a further 31 % below the post-layout TT and below the SS corner; the post-layout ring would need about 0.97 V to match. Candidates: (a) wiring *resistance*, not extracted (the extraction is capacitance only); in a ring of `1v` inverters with 4× fingers this is usually a few percent, not tens; (b) supply and ground drop: RO13 slows 4 % when RO101 runs, which at 12 % per 100 mV means about 40 mV lost for the 0.5 mA of the other ring — some 80 Ω of shared impedance, large for a board, and suggesting a shared on-chip path (both rings return through the core VSS); a 230 mV drop for RO13's own current would need several hundred ohms, so the supply alone does not close the gap either; (c) the model corner, with the SS corner already 10 % faster than the chip. Each is plausible, none is measured, and together they are at the limit of what they can explain. RO13 is therefore *not yet justified*: ROVDD/RO2VDD must be measured at the jumpers with each ring off, alone and both on, and the ring swept from a bench supply.

RO101: not justified by any parasitic. RO101 runs 4.9× faster than its post-layout fundamental, 37 ps per stage for one edge in 101 stages, which this process cannot do (RO13 shows about 290 ps per stage on the same chip). The measured value matches instead a ring with about seven circulating edges: scaled by RO13's factor (0.69) the fundamental would be 19.2 MHz and the measurement is 7.05× that — or, equivalently, a single edge in a loop of about 14 stages, almost RO13's 13. The numbers on the bench support "something short" more than "something slow": $f_{13}/f_{101} = 0.97$ instead of 6.93, a duty cycle of exactly 50 %, and the same frequency after every restart, including the footer-off restart that launches a single edge. In simulation the extracted 101-stage ring does not keep a harmonic mode: an all-zero start begins with 3 edges (4 without wiring capacitance) that merge within 0.75 µs, ending at the fundamental. The extracted cell also has a single 101-stage loop. So either:

- the chip keeps re-creating edges until about seven circulate — the `extra_load` node, slowed by its pad (587 Ω and ESD capacitance), is the first suspect: a slow edge with noise can switch the next stage twice; or
- a short loop exists on silicon that is not in the extracted cell (a defect, or a path outside the cell); or
- the DUT stage is not behaving as designed, which the dead `DUT_Header` control also suggests. One mechanism worth testing: the `Drain_Force`/`Drain_Sense` transmission gates are tied to 1.2 V on the bench; any conduction through them in oscillation mode pulls the DUT output up independently of its header, which would explain why the header has no effect and would distort the DUT edge.

Note that the full-chip LVS matched with the RO netlist flattened, so a wrong chip-level connection of `R[3]`/`R[11]` to `DUT_Header` is unlikely unless it hides in the parts the custom deck ignores. In short, **the RO101 discrepancy is a functional anomaly, not a modelling error**, and it cannot be blamed on the missing extraction.

Next measurements. (1) Multimeter on ROVDD/RO2VDD (JP3, JP4) in every state. (2) Sweep ROVDD from the bench supply: in a harmonic mode RO101 jumps by discrete ratios (7, 5, 3, 1 edges), a short loop follows RO13 smoothly. (3) Scope on `ro_101_ex` (J1 pin 28) with a short ground: several pulses per period would confirm extra edges. (4) Leave `df`/`ds` floating or tie them to ground and repeat the header-off test. (5) Extract the chip around the RO macros (top-level routing of the control bits and of VSS).

8 SPI: the read-back failure

8.1 Symptom

Writes reach the registers (the ROs obey them), but MISO never returns the expected data: neither the constant registers `0xAA`/`0x55` nor the echo of the written registers.

8.2 Cause: one identifier

In `digital/rtl/spix/spi.v` the read multiplexer is indexed by the *write* address:

```
assign addr = sft_reg[numbit_waddress-1+sword:sword]; // Full handshake
assign raddr = sft_reg[numbit_raddress-1:0]; // When the handshake is on addr
...
assign bus_rd = RDA[addr]; // taped out -- should be RDA[raddr]
```

The read word is captured into `bus_cap` after only five bits of the frame (`encap = sync[numbit_address+2]`), when the address sits in the low bits of the shift register, i.e. in `raddr`. `addr` points 8 bits higher, where at that moment the shift register holds bits of the *previous* frame (after five shifts, its data bits [5:3]). The multiplexer therefore selects a stale index: for the zero-padded read frames this is index 0, the ADC-result register, and indices 6 and 7 do not exist (`inputs=6`). The output shift register, `DOUTNOZ` and the `miso` pad are correct; only the selected word is wrong.

A second error compounds it at the top level: `chip/rtl/asicone_202508.v` builds the read word from `RESULT[0..4]`, an identifier that is never declared (the ADC outputs are `pad_adc_result_*_c2p`). Verilog's implicit nets let it through synthesis, PnR and LVS as an undriven 5-bit net, so register 0 is floating on silicon. Together: almost every read returns the undriven register 0, which matches the dead MISO.

8.3 Confirmation

`report/sim/run.sh` reads the SPI with 21-clock frames and the read map of the chip, with the RTL as taped out and with the one-line fix:

```
== as taped out == with RDA[raddr]
read addr 2 -> xx read addr 2 -> aa
read addr 3 -> 00 read addr 3 -> 55
read addr 4 -> 00 read addr 4 -> 06 (R0 echo)
read addr 5 -> 00 read addr 5 -> 3c (R1 echo)
```

(`xx`: out-of-range index 7 selected by the data of the preceding write.)

8.4 Why it escaped

The module testbench (`spi_tb.v`) still connected a port `DOUT` that had been split into `DOUT_DAT`/`DOUT_EN`; it no longer elaborates, so the read path was not simulated after that change, and the full-chip verification was DRC and LVS only, which compare layout with netlist and cannot see a logic error. A third issue is architectural, not a typo: the SPI reset is the `adc_rst` pad, active low for the SPI and active high for the ADC, so the RO configuration is cleared whenever the ADC runs: the ADC and the ROs cannot be operated at the same time.

9 Errata and lessons

#	Issue	Impact	Fix for the next chip
1	SPI read mux uses `addr` instead of `raddr`	MISO dead	One-line RTL fix; read regression
2	Undeclared `RESULT` in the top level	ADC result not readable by SPI	`` `default_nettype none ``; lint (Verilator)
3	SPI reset shared with `adc_rst`, opposite polarity	ADC and ROs exclusive	Separate reset pin or a register bit
4	DAC plate wiring differs per bit, halves not mirrored	ENOB 3.8 instead of 4.7	Nested pattern, rail reassignment, mirrored halves (done in `saradc_auto`)
5	No extraction at tapeout	Layout effects unseen	Split KLayout-PEX flow now available; make it a sign-off step
6	RO101 4.9× too fast, `DUT_Header` without effect	RO101 unusable for its purpose	Open: measurements of Section 7.2
7	RO13 31 % slow	Calibration unclear	Measure supplies; separate RO grounds
8	52 `Sdiod` DRC errors in IHP I/O	None seen	Follow IHP I/O updates
9	`ptap1` ignored in LVS; added `PolyRes`	Reduced LVS coverage	Upstream deck fixes
10	Liberty copied from `sky130`	Timing nominal	Characterize with extraction
11	Board: LMV358 front end, references undecoupled	HF ENOB, possible INL	FDA front end, decoupling (`pcbfix`)

Table 11: Errata.

The common thread is verification coverage under schedule pressure: everything that DRC and LVS check was right on silicon (pad ring, supplies, connectivity of every block); everything that needed a functional simulation of the integrated design (the SPI read path, the top-level read word, the reset sharing) or a parasitic-aware simulation (the DAC, the rings) was not checked, and that is where the chip fails. A lint pass with implicit nets disabled and a top-level RTL testbench that reads every register would have caught issues 1–3 in minutes.

10 Reproducing this report

Chip `asicone_sg13g2` (`chip/`, `digital/`, `ro_reliability/`, `saradc_auto/`); submission copy in `TO_Sep 2025/asicone_202509`. Note that `saradc_auto/pnr/outputs` now holds the hierarchical ADC; the taped-out ADC (`saradc_auto/*/outputs.tapeout`, git-ignored) is archived in the release `tapeout-T594`: `https://github.com/ckdur/saradc_auto/releases/tag/tapeout-T594` (mirror: `https://gitlab.symbioticeda.com/ckdur.iso/saradc_auto/-/releases/tapeout-T594`).

ADC bench `RISCVConsole/software/mso`: `adc_sweep.sh`, then
`make -C report RESULTS=../results/sweep_20261022`.
Simulation: `saradc_auto/sim` (`char.sh`), extraction `saradc_auto/pex/split.py`.

RO bench `RISCVConsole/software/ro`: `ro_run.sh`, then
`make -C report RESULTS=../results/ro_20261029`.
Simulation and extraction: `ro_reliability/sim` (`ro_sim.py`, `pex/run_all.sh`).

Bench data A copy of both result directories (raw captures, CSV, logs; RISCVConsole `c32ae7a`) is in `report/data/adc_sweep_20261022` and `report/data/ro_20261029`; copy one back to `results/` to re-run its report.

SPI `report/sim/run.sh` (iverilog).

Board `pcbs/IHP130_ASICONE_ADP_QFN48_0p5Q_Adapter` (KiCad, BOM, CPL).

This document `latexmk -pdf report.tex` in `report/`.

Acknowledgements

IHP for the OpenMPW tapeout; Symbiotic EDA, which sponsored the tapeout and uses the design to evaluate open-source tools; The University of Tokyo, which uses it for research.

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